

Digital Investment Behavior: Insights from Gen-Y and Gen-Z

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Abstract—Digital investment services have changed investor behavior from conventional to more practical and flexible. Investors use investment applications that feel safe and friendly to operate. This study aims to identify the UTAUT model in understanding the behavior of Generation Y and Z investors in using investment applications. An online survey was conducted among 294 investors from Y and Z generations in Indonesia to answer the research objectives. The collected data were analyzed using multivariate-Structural Equation Modeling (SEM) based on the covariate. The results of this study have proven that performance expectancy, effort expectancy, facilitating condition, and social influence have a positive and significant effect on behavioral intention. Likewise, facilitating conditions and behavioral intentions positively and significantly affect user behavior. The four factors in the UTAUT model become determinants for investors in making decisions on using investment applications. In conclusion, understanding consumer behavior in adopting technology is crucial for companies developing technology.

Keywords—UTAUT, Digital Investment, Gen Y and Z, CB-SEM, Indonesia

I. INTRODUCTION

The financial services sector in Indonesia has undergone significant changes due to the presence of digital product innovations. These changes force companies to adopt new technologies to survive in a competitive market. Almost all companies are racing to create digital platforms in order to be able to provide services flexibly to all of their customer segments, including investment services. The investment sector is an example of a sector that has undergone drastic changes due to innovation. Innovation has changed how companies offer investment services with new standards and ways of serving every investor.

In Indonesia, digital investment platforms began to grow in the mid-2000s. In 2021, several digital-based investment startups will be present to enliven the investment markets competition, such as Bareksa, Bukalapak, Bibit, Pluang, Nanovest, and others [1]. Based on Central Bureau of Statistics (BPS) 2020, the population of gen-Z reached 27.94%, and gen-Y was 25.87% of the total population of Indonesia in 2020 [2]. The 2021 digital literacy survey results

show that 60% of gen-Z belongs to a group with a high level of digital literacy, while gen-Y has a high level of digital literacy of 54%. This condition explains that the younger generation is quickly exposed to and connected to digital technology. Therefore, both generations are profitable target segments for investment companies.

Gen-Y and Gen-Z have something in common regarding the use of technology. The difference is that Gen-Y prefers communicating via text or voice, while Gen-Z prefers video communication [3]. Therefore, prospective investors from Gen-Y and Gen-Z tend to experience changes in behavior that are more adaptive to information technology.

Previous studies have proven that Gen-Y and Gen-Z are profitable segment groups for digital-based companies because of their high intention to continue adopting technology. Example, Windasari, et al. [4] examines Gen-Y and Gen-Z regarding their experience using digital bank services. Lee and Gan [5] discussed e-wallet adoption among Gen-Y and Gen-Z in Malaysia. Puiui, et al. [6] studied the Gen-Z group in adopting mobile commerce, and Hasan, et al. [7] examined cryptocurrency adoption among Gen-Y and Gen-Z groups.

Referring to the results of previous research and the high intention of Gen-Y and Gen-Z in adopting technology, this study aims to investigate the behavior of adopting technology in investment services among Gen-Y and Gen-Z groups in Indonesia using the UTAUT conceptual framework.

II. LITERATURE REVIEW AND RESEARCH HYPOTESIS

A. UTAUT

Many theories are used to answer challenges for researchers, especially in the field of information systems and management, such as the theory of diffusion of innovation (DoI), IDT, and models of using computers to explain the adoption of IS/IT separately [8]. UTAUT comprehensively examines all eight dominant technology adoption models to overcome the limitations of technology acceptance theory [9]. UTAUT's development emphasizes the concept of acceptance of previous technologies. UTAUT emerged as the most popular theory in understanding the adoption of mobile-based payment technology [8]. The UTAUT model encapsulates

four core constructs: performance expectations, effort expectations, social impact, and facilitating conditions. Modeling behavioral intentions and attitudes towards the use of technology in stages to analyze behavior mechanisms [10].

B. The relationship between performance expectancy and behavioral intention

Performance expectancy is a significant predictor of behavioral intention to use a mobile-based payment system because it relates to performance interests. The higher the efficiency and performance of mobile payments, the greater the behavioral intention to use the application [11]. These findings are in line with the research of Lukito and Ikhsan [12], which proves that e-commerce performance is a stimulus for individuals to make transactions again. Developing technology that prioritizes performance will encourage individual intentions to adopt it. Jadir, et al. [13] explained that performance expectancy plays an essential role in shaping behavioral intentions. Individuals will be more inclined to use mobile banking if they feel that technology will provide benefits in conducting banking activities.

Khechine, et al. [14] research proved that performance expectancy is not the main predictor of behavioral intention. It happens because individuals have other options to meet their needs rather than using existing information systems. There is an assumption that new information technology only sometimes has better performance than previous technology, so it impacts personal reluctance to use the new technology [15]. Based on the research gap, the hypothesis is formulated as follows.

H1: Performance expectancy influences individual behavioral intention in using digital investment services.

C. The relationship between effort expectancy and behavioral intention

Effort expectancy is related to information technology's ease of use and complexity [16]. The existence of ease and complexity in operating technology encourages individuals to use mobile payments [11]. Individuals can find M-banking easily and practically, increasing their desire to carry out banking transactions [17]. These results align with research by Dewi, et al. [18]; individuals only need to make more effort to operate e-wallets to shop because it has straightforward steps.

Al-Azawei and Alowayr [19], in an empirical study, explained that effort expectancy did not confirm the relationship with students' behavioral intentions to use learning technology. Student readiness is outstanding in using mobile-based learning information systems, regardless of the individual effort required. Students are used to using such technology in everyday life, so they have had much experience [19]. The results of other studies prove that the main predictor of behavioral intention is only sometimes effort expectancy. It is because individuals find it difficult when using electronic document systems. Based on the research gap, the hypothesis is formulated as follows.

H2: Effort expectancy influences individual behavioral intention in using digital investment services.

D. The relationship between social influence and behavioral intention

Social influence is a view that can be positive or negative from the information technology used by a group. Social groups consist of several individuals who provide mutual

references to either fellow group members or other individuals to gain recognition and acceptance [20]. Community members will be influenced by individuals around them, especially those who are very trusted and essential to them [16]. Previous research has shown that individual behavioral intentions to transact using mobile payments are influenced by social influence [21, 22]. The same finding is shown by Trinh, et al. [23], which state that behavioral intention to use credit cards is influenced by social influence, namely people who are considered essential such as family, peers, and co-workers. Social influence directs individual perceptions because significant pressure from others determines the decision to use mobile banking [13].

Different findings show that individual concerns about social influence impact hesitation to use mobile learning [19]. In addition, the voluntary use of e-participation makes social influence unimportant for individuals. The presence of new technology that is rarely used will be deemed not to have affected the norms of the social environment, thereby not encouraging individual intentions to adopt the technology [24]. The results of research by Widodo, et al. [25] showed that social influence is not the most potent predictor that influences individual behavioral intentions to use e-wallets. Raza, et al. [17] also prove that social influence has no impact on behavioral intentions to use mobile banking because customers still have other options for transacting with internet banking. Based on the research gap, the hypothesis is formulated as follows.

H3: Social influence affects individual behavioral intention in using digital investment services.

E. The relationship between facilitating condition with behavioral intention and use of behavior

Facilitating conditions describe the extent to which individuals feel that the existence of technological infrastructure is an encouragement to use it in completing a job [16]. Individuals with the ease of operating information systems will benefit in the speed of completing their work so that they are encouraged to use it for a long time [24]. The availability of technological devices will stimulate individual intentions to transact using a digital wallet [25]. The same results were shown by Patil, et al. [8] that individuals with mobile devices tend to use mobile payments in transactions. The results of this study are in line with Jadir, et al. [13] which state that the availability of complete infrastructure makes it easier for individuals to carry out banking transactions through mobile banking so that there is an intention to use this technology.

The results of different studies revealed by Shaw and Sergueeva [26], facilitating condition is not the primary predictor in influencing behavioral intention. It is because the infrastructure that does not meet the needs will not encourage individuals' intentions to adopt it. There is infrastructure, but more than its function is needed to solve problems at work, so there is no incentive to use the infrastructure again. Wei, et al. [15] stated that facilitating conditions do not confirm behavior intention. Incompatible technological devices make individuals reluctant to adopt them. Individuals feel they need more information about how to use technology, so it becomes one of the reasons that individuals do not want to use it. Based on several differences in research findings that previous experts have carried out, the researcher proposes the following hypothesis

H4: Facilitating conditions affect behavioral intention in using digital investment services.

H5: Facilitating conditions affect user behavior in using digital investment services.

F. The relationship between behavioral intention and use behavior.

Behavioral intention plays a vital role in use behavior [16]. Individual willingness influences the intention to use technology. The results of an empirical study conducted by Raza, et al. [17] showed that features in mobile payment applications that are easy to understand have an impact on customers voluntarily using them. These results are supported by the findings of Palau-Saumell, et al. [27], which show an individual desire to use mobile applications in ordering food at restaurants. Individuals provide recommendations to others so that an intention arises to use mobile payments in transactions [21]. The findings of Jadil, et al. [13] corroborate the results of previous studies; Behavioral intention is a predictor of mobile banking usage behavior. The results of Khechine, et al. [14] stated that behavioral intention does not confirm the behavior of using social learning systems. It happens because the individual does not desire to reuse the information system. Based on several differences in research findings that previous experts have carried out, the researcher proposes the following hypothesis.

H7: Behavioral intention influences the use of digital investment services.

III. METHOD

A. Research Design

This study uses the UTAUT conceptual model to investigate the behavior of using investment applications in Gen-Y and Gen Z. Therefore, a verification approach is used to answer the six research hypotheses. Investment application users in Indonesia comprise the population, while the sub-population is Gen-Y and Gen-Z investors. The determination of sample size in the Gen-Y and Gen-Z groups of 230 investors (23 indicators $\times$ 10) [28] was determined purposively. The data collection period for three months: March–May 2022, using an online questionnaire distributed via social media platforms: Instagram, Twitter, and Facebook. During the data collection period, 249 responses were obtained, and all the data collected was deemed fit for analysis.

B. Measurement

Measurement for each variable refers to previous studies using a semantic differential scale point of 5 (1 = strongly disagree – 5 = strongly agree) (Table I).

TABLE I. MEASUREMENT OF VARIABLE

Variable	Indicator
Performance expectations [29-32]	1. Simplify Work 2. Assist in completing job 3. Increase productivity 4. Improve Performance
Effort expectations [29-32]	1. Easy to learn 2. Easy to implement 3. Easy to understand 4. Easy to operate
Social Influence [29-32]	1. Influence of partners 2. Competitor influence 3. Organizational support 4. Social status
Facilitating conditions	1. Capabilities 2. Knowledgeable

Variable	Indicator
[29-32]	3. Compatible 4. Availability of references
Behavioral Intention [29-32]	1. Interested in using 2. Motivated to use 3. Choose using 4. Increasing the intensity of using digital investment services use
Use behavior [29-32]	1. Simplify business activities 2. Simplify virtual management 3. Simplify the transaction process

C. Structural Equation Modeling

This study uses covariance-based structural equation modeling (CB-SEM). In CB-SEM, there are two measurements: model measurements and structural models [28]. Model measurements use confirmatory factor analysis (CFA) and average variance extracted (AVE) to test convergent validity. Measurement items are declared valid if the loading factor (LF) and AVE values are > 0.5 [28]. At the same time, the discriminant validity test uses the Fornell-Lacker criterion method [33]. Reliability testing uses Construct Reliability (CR) with a CR value limit of > 0.7 [28].

Before testing the structural model, the goodness of fit model (GoF) criteria must be met. There are three measures used in CB-SEM: absolute, incremental fit, and parsimony [28]. Each of these sizes has minimum criteria that must be met—structural model testing to answer the hypothesis concerning the sign value < 0.05 .

IV. RESULT AND DISCUSSION

A. Demographic

This study consisted of two generations: Gen Y and Z. Based on Table II, respondents with Gen Y (23-38 years) were 174 (59.18%), and Gen Z (19-22 years) 120 (40.82%) with male sex by 61% and women 39%. The average respondent's last education was a bachelor's degree (46%) and master's degree (35%), and they worked as a private employee (33%) and government employees (32%). Respondents have used investment applications since 2021, as many as 56% and 2022 as many as 44%.

TABLE II. DEMOGRAPHIC

Demographic	Freq	%
Gender		
Male	180	61%
Female	114	39%
Age		
19 – 22	120	41%
23 – 26	50	17%
27 – 30	69	23%
31 – 34	35	12%
35 – 38	20	7%
Education Level		
Ph.D	17	6%
Magister	102	35%
Bachelor	135	46%
Higher School	40	14%
Job		
Student	40	14%
Government employees	95	32%
Private employees	98	33%
Entrepreneur	36	12%
Professional	25	9%
Using Apps, since		
2021	165	56%
2022	129	44%

B. Result

Applying the CB-SEM method requires that the research data is normally distributed. Table III presents the results of the normality test using the Kolmogorov-Smirnov technique. As a result, the Monte-Carlo sign values for the behavioral intention and use behavior variables are 0.300 and 0.069 > 0.05. Therefore the data is normally distributed.

TABLE III. NORMALITY TEST

One-Sample Kolmogorov-Smirnov Test	BI	UB
Monte Carlo Sig. (2-tailed)	0.300	0.069

Convergent validity testing refers to the Loading Factor (LF) and Average Variance Extracted (AVE) values. At the same time, reliability refers to the Composite Reliability (CR) value. Based on Table IV, all indicators for each UTAUT variable produce LF and AVE values > 0.5 and CR values > 0.7. Therefore, the data questionnaire has met the valid and reliable requirements.

TABLE IV. CONFIRMATORY FACTOR ANALYSIS (CFA)

Variabel	Item	LF	CR	AVE
Performance Expectancy	PE1	0.721	0.822	0.536
	PE2	0.745		
	PE3	0.705		
	PE4	0.757		
Effort Expectancy	EE1	0.845	0.843	0.575
	EE2	0.800		
	EE3	0.742		
	EE4	0.629		
Social Influence	SI1	0.721	0.813	0.521
	SI2	0.721		
	SI3	0.770		
	SI4	0.673		
Facilitating Condition	FC1	0.728	0.815	0.527
	FC2	0.794		
	FC3	0.749		
	FC4	0.621		
Behavioral Intention	BI1	0.742	0.831	0.552
	BI2	0.807		
	BI3	0.761		
	BI4	0.655		
Use Behavior	UB1	0.729	0.773	0.531
	UB2	0.721		
	UB3	0.736		

Discriminant validity testing uses the Fornell-Lacker criteria. In Table V, it can be seen that the AVE root value is greater than the correlation between other variables, so the measurement model has good discriminant validity.

TABLE V. DISCRIMINANT VALIDITY – FORNELL LACKER

	PE	EE	SI	FC	BI	UB
PE	0.732					
EE	0.446	0.758				
SI	0.484	0.577	0.722			
FC	0.437	0.516	0.712	0.726		
BI	0.539	0.609	0.710	0.723	0.743	
UB	0.441	0.578	0.703	0.719	0.739	0.729

CB-SEM analysis has a good measurement model if it meets predetermined criteria. Based on Table VI, both the CFA measurement model and the structural model have produced good measurement values. Only the Chi-Square and Prob sizes are not fulfilled. Therefore, the CFA and structural models meet the Goodness of Fit (GoF) criteria.

TABLE VI. GOODNESS OF FIT MODEL

GoF index	Structural Model	CFA	Decision
Absolute Measures			
Chi-square	305.06	297.67	Poor
DF	218	215	
Probability	0.00009	0.00016	Poor
GFI	0.917	0.920	Good
RMSEA	0.037	0.036	Good
CI of RMSEA	0.03–0.05	1.29–1.60	
RMR	0.0175	0.017	Good
SRMR	0.041	0.041	Good
Normed chi-square	1.399	1.385	Good
Incremental Fit Measures			
NFI	0.977	0.980	Good
NNFI	0.992	0.990	Good
CFI	0.993	0.990	Good
RFI	0.974	0.970	Good
Parsimony Measures			
AGFI	0.895	0.900	Good
PNFI	0.842	0.830	Good

The next step is testing the structural model to prove the hypothesis. Table VII and Figure 1 show that all research hypotheses are accepted because each path coefficient value produces a sig. < 0.05, and the standardized value is positive. It means that effort expectancy, facilitating condition, performance expectancy, and social influence positively and significantly affect behavioral intention. Likewise, facilitating conditions and behavioral intentions positively and significantly affect user behavior. The constructs of effort expectancy, facilitating condition, performance expectancy, and social influence contribute 90.5% in forming a behavioral intention, while facilitating condition and behavioral intention contribute 95.5% in developing use behavior.

TABLE VII. THE RESULT

Hipotesis	STD	T stats	R ²
Effort Expectancy→Behavioral Intention	0.154	2.683	0.905
Facilitating Condition→Behavioral Intention	0.372	2.594	
Performance Expectancy→Behavioral Intention	0.112	2.151	
Social Influence→Behavioral Intention	0.417	2.602	
Behavioral Intention→Use Behavior	0.607	3.941	0.955
Facilitating Condition→Use Behavior	0.393	2.602	

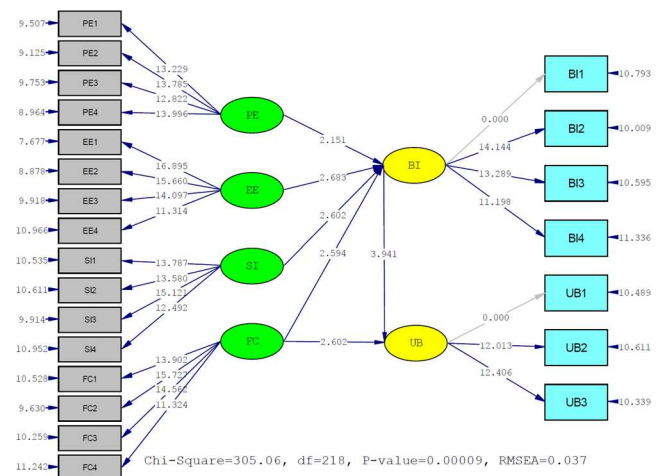


Fig. 1. Stuctural Model

C. Discussion

The study's results prove that the UTAUT conceptual framework can explain the behavior of using digital investment services. In detail, performance expectancy positively and significantly affects the behavioral intention to use digital investment services. This finding is in line with previous researchers who explained that the performance of digital investment services encourages individual intentions to adopt and use them [11, 13]. *Digital investment* is a technological innovation that produces services with good performance. Individuals' intentions will be encouraged to use digital investment services if there are benefits.

Subsequent findings show that effort expectancy positively and significantly affects behavioral intentions in using digital investment services to make transactions. These results are in line with an empirical study conducted by Al-Saedi, et al. [11] explained that the existence of a level of ease and complexity in operating a technology will further encourage individual intentions to use it. This finding is in line with Shaw and Sergueeva [26], who stated that individuals do not need excessive efforts to operate digital investment services.

The results of this study prove that social influence has a positive and significant influence on behavioral intention. These findings are in line with the results of research by Patil Patil, et al. [8], and Jadir Jadir, et al. [13], which show that individual behavioral intentions arise due to encouragement from individuals or groups who have had previous references regarding digital investment services. The influence of business partners, competitors, support, and organizations drives individual behavioral intentions in using digital investment services.

The subsequent results prove that facilitating conditions positively and significantly affect behavioral intentions. This finding is in line with the research of Naranjo-Zolotov, et al. [24] that the existence of technological facilities will make it easier for individuals to complete a job. The availability of technological facilities can stimulate behavioral intentions for individuals to use digital investment services because it will make transactions easier.

The findings show that facilitating conditions significantly positively affect user behavior in digital investment services. These results align with the research of Jadir, et al. [13] namely digital investment services according to individual needs in transactions. Individuals with knowledge and references tend to use digital investment services.

The last finding in this study proves that behavior intention has a positive and significant effect on the user behavior of digital investment services. These results align with research conducted by Raza, et al. [17] that individual interest raises the intention of individuals to use digital investment services when making transactions. Nwankpa and Merhout [30] confirmed that the motivation and desire to reuse will encourage individuals always to use digital investment services in transactions.

D. Conclusion, Limitation and Future Research

The conclusion of this study shows that individual behavioral intentions regarding the use of digital investment services are based on performance expectancy, effort expectancy, social influence, and facilitating conditions. The most significant users of digital investment services are

generation Y and generation Z. The more digital investment service providers in Indonesia need to pay attention to convenience, and social influence, to build an interest in individuals because it will encourage someone's intention to decide to use the service. Digital investment in conducting a transaction. The capability of digital investment services is also a driving force for individual intentions to use them.

This research has limitations that cannot be reached by research. This limitation opens up opportunities for further research. First, only four UTAUT constructs are used to understand the behavior of using investment applications. There are still several factors, such as habit, satisfaction, and others. Second, this research does not examine the role of gender and age in the moderation function. Because Indonesian user is still new to using digital investment services, further researchers can test the linked model with construct extensions to comprehensively understand the behavior of using investment applications.

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